

Practicals Control design applied to Mobility : Project

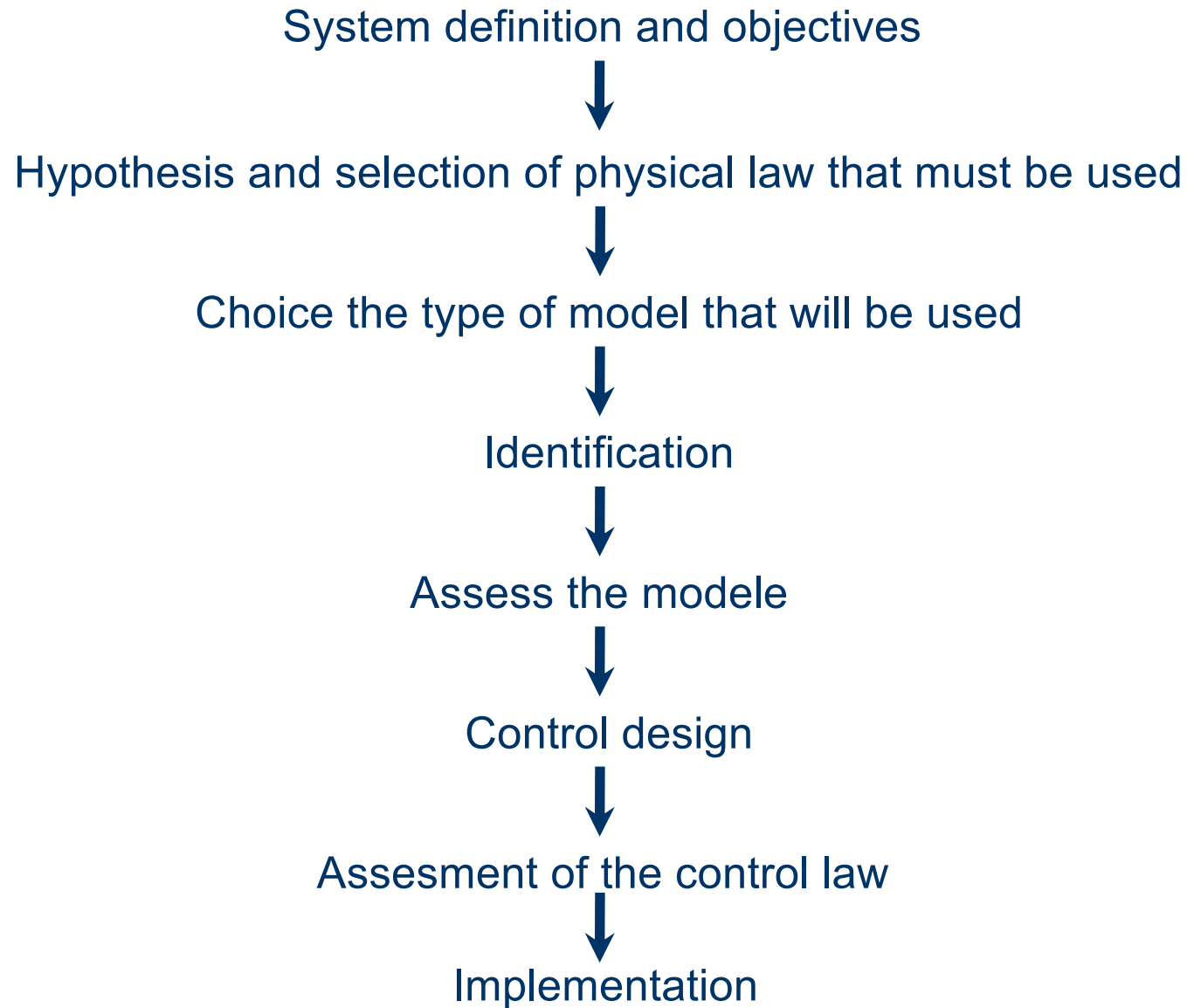
Vehicle's Idle speed control



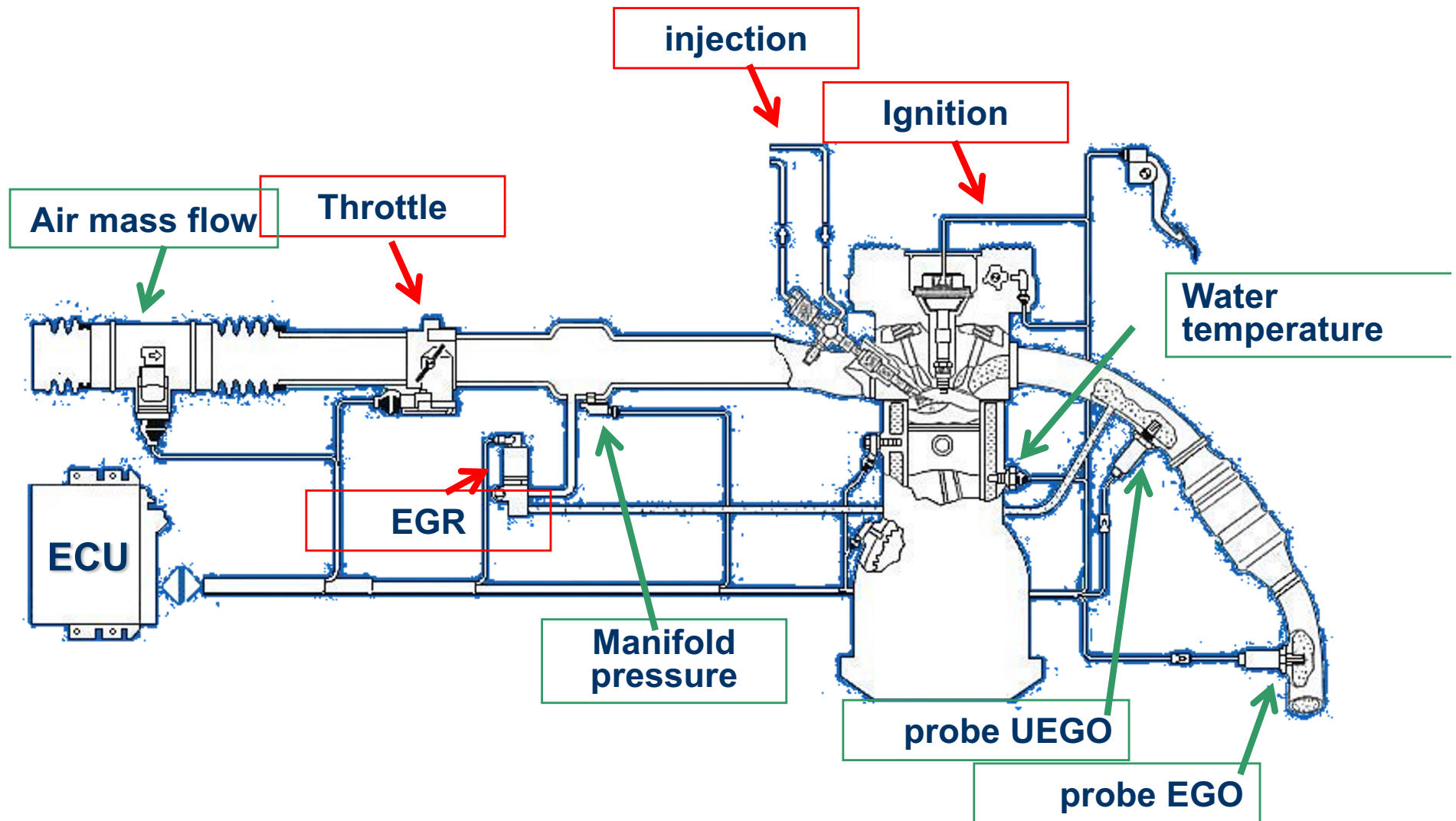
General objective fo this project

- Learning and apply all steps to build a control solution:
 - Problem analysis,
 - Control problem formulation,
 - Modeling,
 - Linearizing the system,
 - Proposed control solutions and Design,
 - Robustness analysis,
 - Tests.
- Slot 1 : Presentation of the project,
- Slot 2 : Presentation of the control part,
- Slot 3 : Control of the system
- Slot 4 : Control of the system
- Slot 5 and 6 : Presentation

Vue générale du TD



Entrées / Sorties du système



La Régulation de ralenti

- Control law specifications
 - Inputs/outputs definition
 - Control law objectives
 - Challenges,
 - Performances,
 - Robustness.
- Control design approach
 - Reference model design
 - Block diagram structure,
 - Submodels definition
 - control law synthesis
 - design
 - simulation

System's inputs and outputs

- 2 inputs and 1 output

- Inputs

- Cyclic opening valve :

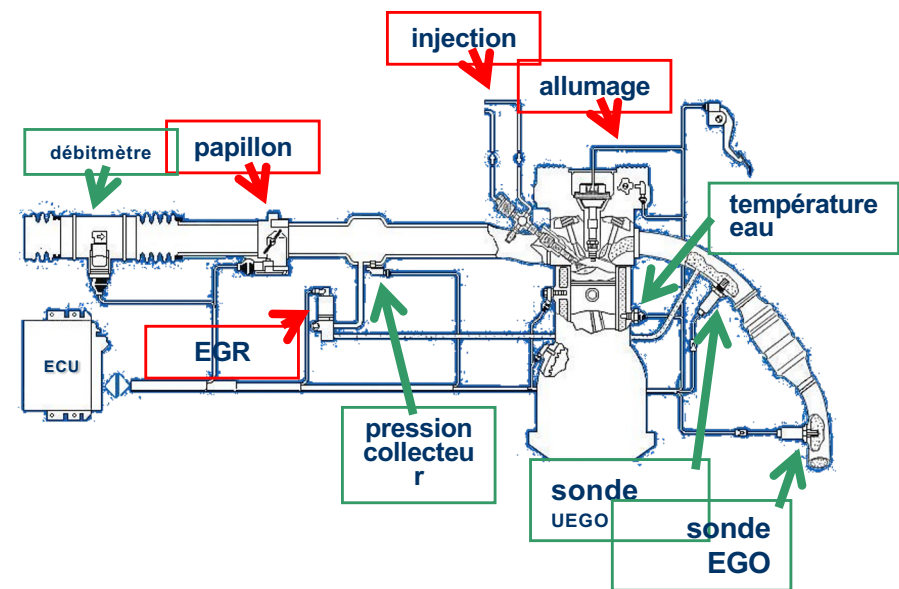
- We control the position of the valve then this position is converted into cross-sectional area [mm²]

- Ignition :

- We control the crank angle degree where the spark is produced then this control input is converted into optimum torque output [%]

- Output

- Speed engine: [tr/mn]



Issues related to idle speed regulation

Control design problem with disturbances rejection

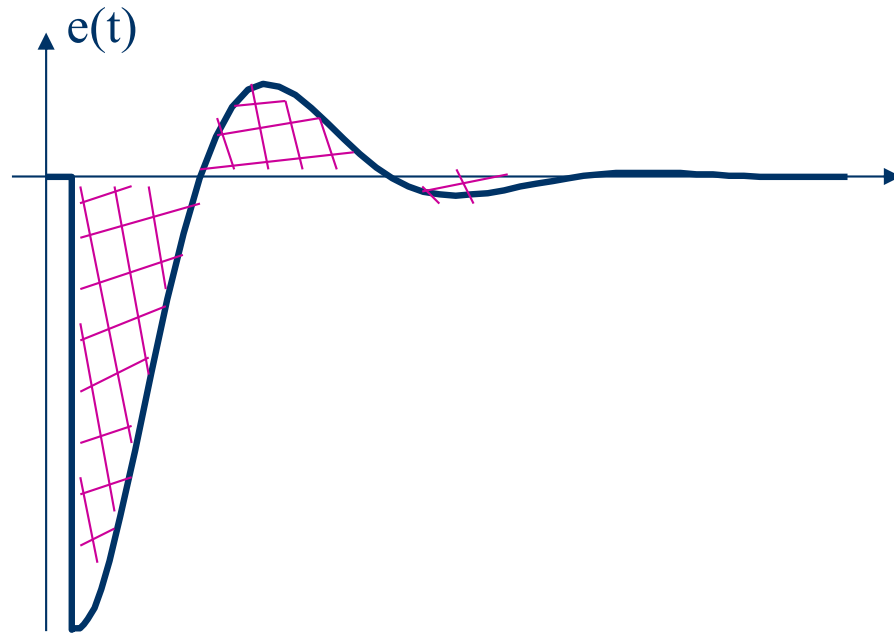
- Output criteria
 - Lower the speed engine setpoint
 - Engine stability : architecture, combustion, efficiency, ...
 - Settling time of the control law.
 - Avoid pumping and vibration :
 - low-frequency oscillations in response to disturbances
 - Return to idle speed.
 - Minimize standard deviation :
 - High-frequency oscillations.
- Constraints on inputs:
 - Cyclic opening valve
 - Low sensitivity to noise.
 - ! Disrupts the control of the composition of the fuel mixture.
 - Ignition advance:
 - Minimize this action
 - No integral action needed
 - ! Reduce torque margin

Question

- How can these issues be expressed in a single criterion $J=?$

TIME RESPONSE SPECIFICATIONS

- Temporal performance can also be expressed through integral functions of the error.



CRITERIA IAE $J_{IAE} = \int_0^{\infty} |e(\tau)| d\tau$

CRITERIA ISE $J_{ISE} = \int_0^{\infty} e^2(\tau) d\tau$

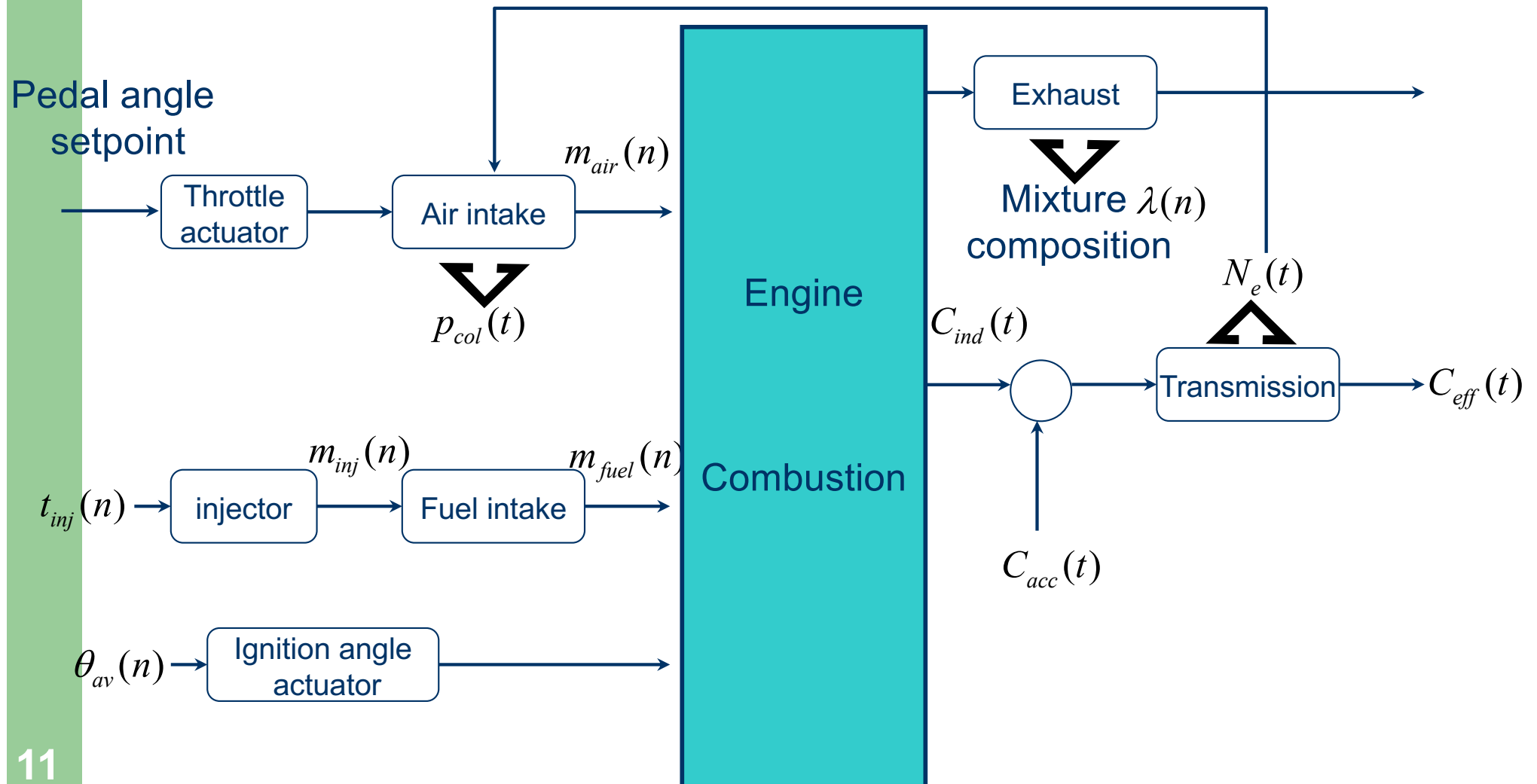
CRITERIA ITAE $J_{ITAE} = \int_0^{\infty} \tau |e(\tau)| d\tau$

QUADRATIC NORM $J_2 = \sqrt{\int_0^{\infty} e^2(\tau) d\tau}$

- Input control level could be taken into:

$$J_{eu} = \sqrt{\int_0^{\infty} Qe^2(\tau) + Ru^2(\tau) d\tau}$$

Block diagram representation



Inputs control values apply

- Fuel injection/ mass air flow control is conducted in three steps :
 - ① States system measurement:
 - state of the engine:
 - speed - N_e
 - Manifold pressure - P_{in}
 - température ...
 - Cyclic opening valve, throttle position
 - ② Inputs control value calculation:
 - Air actuators [mm^2]
 - Injection actuators [T_{inj} , angle]
 - ③ Apply the inputs control values
 - Cyclic opening valve,
 - injection : One round before the admission phase
- Ignition Advance Control
 - ① States engine measurement
 - speed - N_e
 - Operating points (M_{air} , EGR)
 - ② Control value calculation SA
 - ③ Apply at the current TDC.

Question

- the model...

Modelization Hypothesis

Continuous Average Model

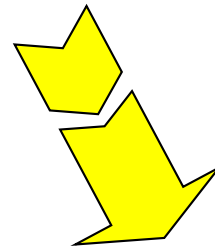
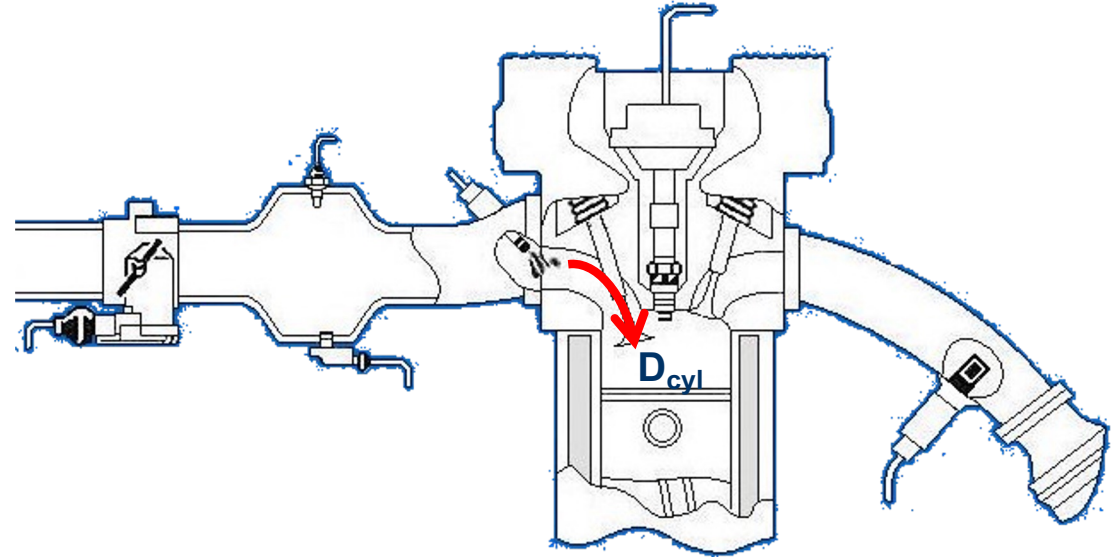
- *Continuous model:*
 - Engine = Hybrid system : Time dynamics + Event dynamics,
- *estimation of the average state variables on each TDC:*
 - Continuous low frequency model
 - average filling/emptying over a TDC
 - Mean model with 4 cylinders
 - undifferentiated torque production between cylinders
 - due to cylinder filling differences
 - Fuel mixture is not taken into account in the production of a couple
 - Consideration of combustion "noise"
- *Pure delays not taken into account*

Problem of air mass

Objective

Determine the amount of air admitted into the cylinder

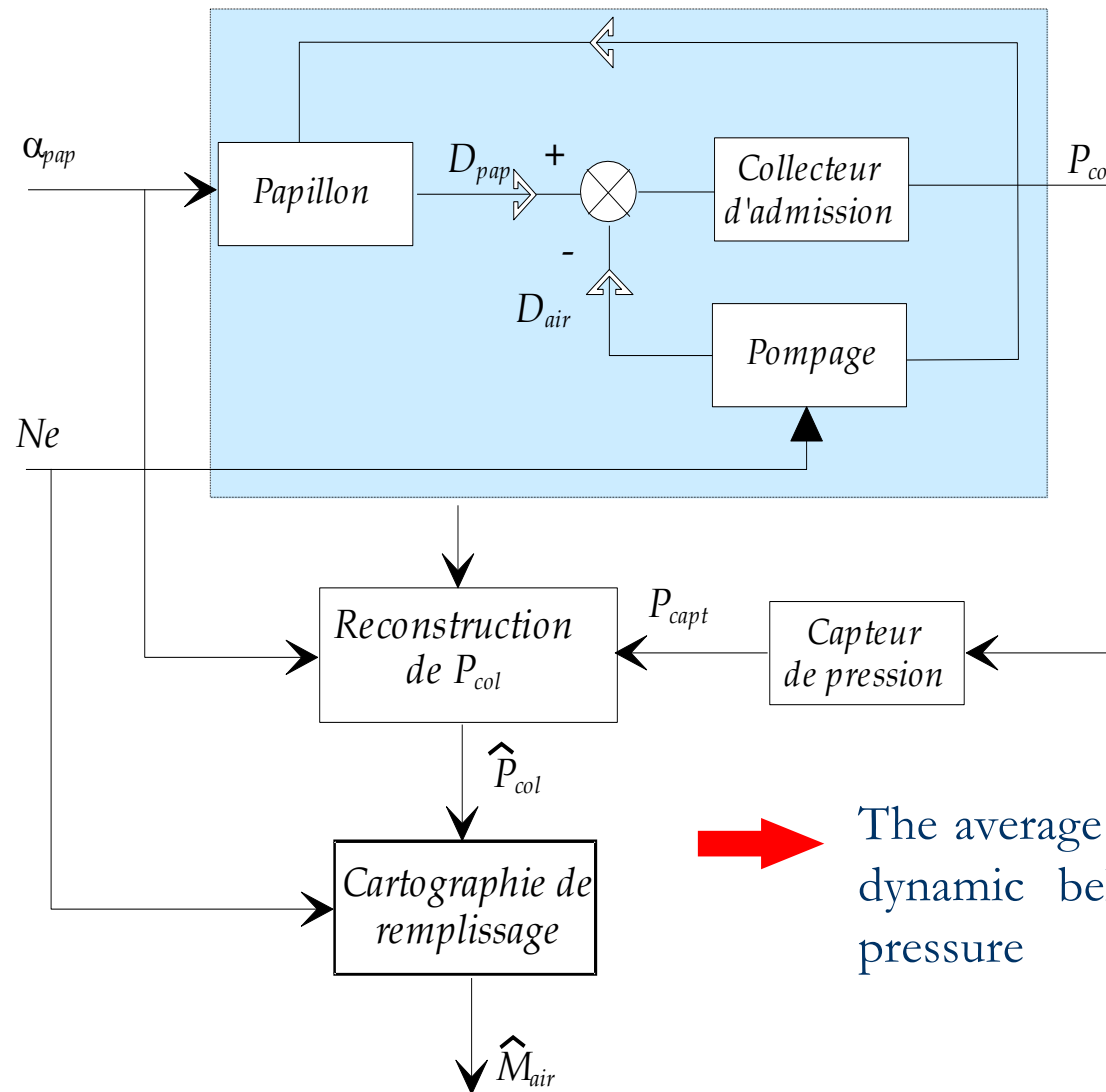
$$m_{air} = \int_{\tau_{soup}} D_{pump}(t).dt$$



Problem

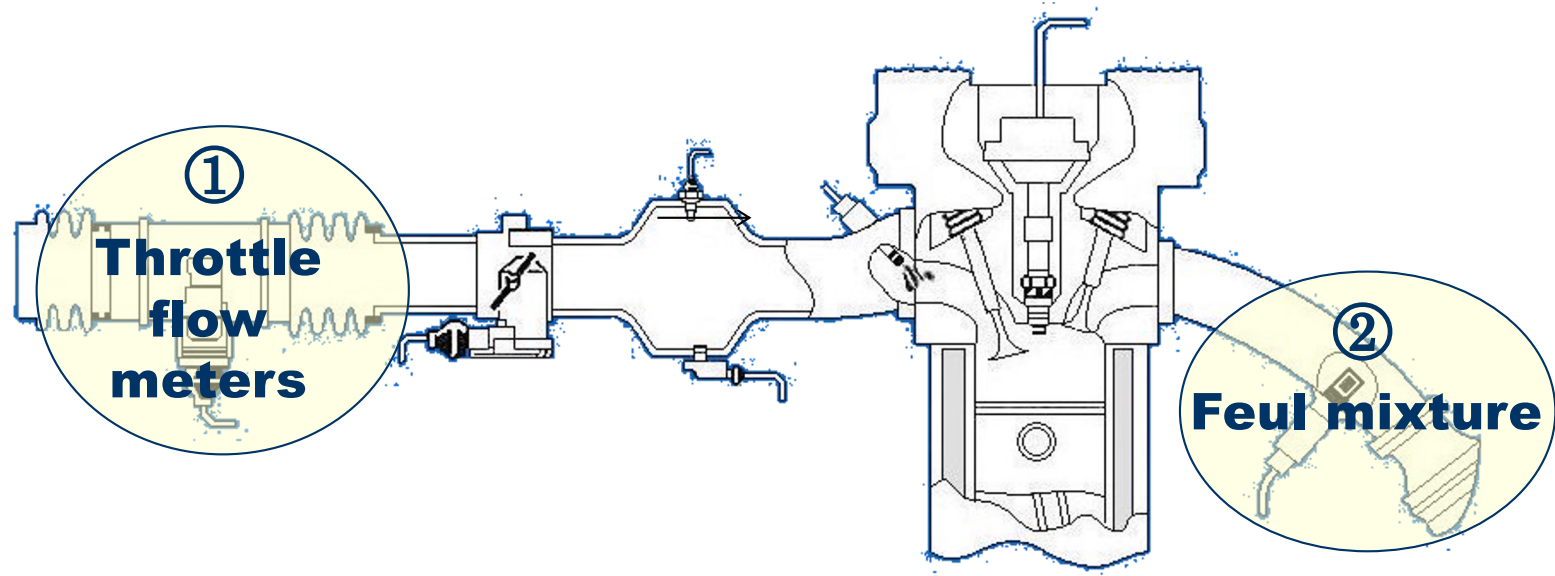
No flow meter to measure airflow at valves

Intake air dynamics - assumption of an average model

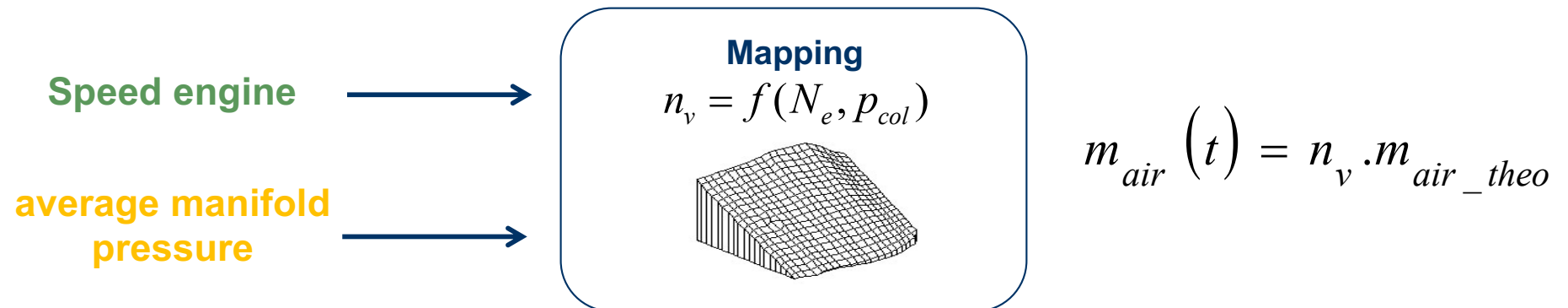


➔ The average airflow at the valves has the same dynamic behavior as that of the manifold pressure

Air mass estimation solutions



Map approach : « Speed Density Method »



Manifold pressure estimation

- Assumptions :
 - ideal gases laws,
 - conservation of mass,
 - uniformity of the quantities observed in the manifold: pressure, temperature,
 - effects of fuel injection on airflow neglected.

$$D_{col} = D_{pap} - D_{cyl}$$

$$D_{col} = \frac{d}{dt} \left[\frac{p_{col} V_{col}}{r T_{col}} \right] \quad \Rightarrow \quad \dot{p}_{col} = \frac{r T_{col}}{V_{col}} (D_{pap} - D_{cyl}) + p_{col} \frac{\dot{T}_{col}}{T_{col}}$$

P_{col}	[Pa]	pression dans le collecteur d'admission.
D_{col}	[kg.s ⁻¹]	débit massique dans le collecteur d'admission.
D_{pap}	[kg.s ⁻¹]	débit massique d'air passant par le papillon.
D_{air} / M_{air}	[kg.s ⁻¹] / [g]	débit massique / masse d'air aspiré par le moteur.
$r = \frac{R}{M}$	[J.kg ⁻¹ . K ⁻¹]	constante des gaz parfait $R = 8.314$, M = masse molaire de l'air $r = 287$
T_{col}	[K]	température du collecteur d'admission.
V_{col}	[l]	volume du collecteur d'admission.

Manifold pressure estimation

- Assumptions :

- The heat transfer in the intake manifold varies slowly with respect to the pressure evolution

➔ thus the term derived from temperature can be neglected in front of the other terms

$$\Rightarrow \dot{p}_{col} = \frac{rT_{col}}{V_{col}} \left(D_{pap} \left(\alpha_{pap}(t), p_{col}(t) \right) - D_{cyl}(N_e(t), p_{col}(t)) \right)$$

p_{col} [Pa] pression dans le collecteur d'admission.

D_{col} [kg.s⁻¹] débit massique dans le collecteur d'admission.

D_{pap} [kg.s⁻¹] débit massique d'air passant par le papillon.

D_{air} / M_{air} [kg.s⁻¹] / [g] débit massique / masse d'air aspiré par le moteur.

$r = \frac{R}{M}$ [J.kg⁻¹. K⁻¹] constante des gaz parfait $R = 8.314$, M = masse molaire de l'air
 $r = 287$

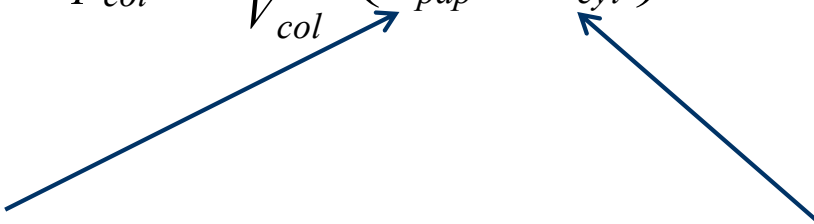
T_{col} [K] température du collecteur d'admission.

V_{col} [l] volume du collecteur d'admission.

$$\Rightarrow \dot{p}_{col} = \frac{r}{V_{col}} \left(D_{pap} \left(\alpha_{pap}(t), p_{col}(t) \right) T_{pap} - D_{cyl}(N_e(t), p_{col}(t)) T_{cyl} \right)$$

Estimateur/prédicteur de pression collecteur

L'estimateur/prédicteur est basé sur le modèle « Filling and Emptying »

$$\frac{\Sigma}{p_{col}} = \frac{r T_{col}}{V_{col}} (D_{pap} - D_{cyl})$$


Solution 1

estimation du débit au papillon par la **relation de Barré de Saint-Venant**

$$D_{pap}(a_{pap}, p_{col}) = A(a_{pap}) \cdot f(p_{col})$$

Solution 2

mesurer le débit au niveau du papillon

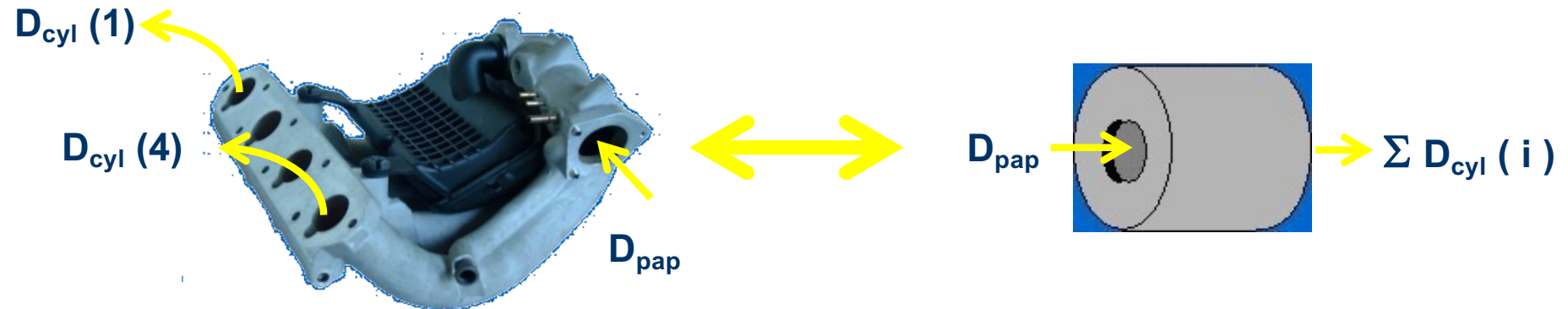
estimation du débit aux soupapes à partir de la **cartographie de remplissage**

$$D_{cyl} = h_{vol}(N_e, p_{col}) \frac{V_{cyl}}{2} \frac{N_e}{60} \frac{p_{col}}{r T_{col}}$$

représente le fonctionnement moyen du moteur sur 1 PMH

Calculation of intake valve flow by the model approach

Manifold model « Filling and Emptying Model »



- Conservation of mass
- No heat exchange (EGR, back flow)

Manifold pressure evolution as a function of flow rates: Integrating system

$$\frac{\sum}{\dot{p}_{col}}(t) = \frac{r T_{col}}{V_{col}} (D_{pap}(a_{pap}(t), p_{col}(t)) - D_{cyl}(N_e(t), p_{col}(t)))$$

Average Valve Flow

$$D_{cyl} = h_{vol}(N_e, p_{col}) \frac{V_{cyl}}{2} \frac{N_e}{60} \frac{p_{col}}{r T_{col}}$$

Throttle flow calculation

$$D_{pap} = A(a_{pap}) \cdot f(p_{col})$$

- Cross-section Area model :

- ① Circular section

$$A(\alpha_{pap}) = C_d \left[(\cos \alpha_0 - \cos(\alpha_{pap} + \alpha_0)) \cdot S_{pap} + S_{fuite} \right]$$

- ② Elliptical section

$$A(\alpha_{pap}) = C_d \left[\left(1 - \frac{\cos \alpha_{pap}}{\cos \alpha_0} \right) \cdot S_{pap} + S_{fuite} \right]$$

$$\left\{ \begin{array}{ll} C_d & [] \quad \text{coefficient de débit} \\ \alpha_0 & [\text{rad}] \quad \text{angle de fermeture} \\ S_{pap} & [\text{cm}^2] \quad \text{section} \\ S_{fuite} & [\text{cm}^2] \quad \text{section de fuite} \end{array} \right.$$

- Barré St Venant:

- ① Subsonic mode

$$\frac{P_{col}}{P_{amb}} > \left(\frac{\gamma + 1}{\gamma} \right)^{\frac{\gamma}{1-\gamma}}$$

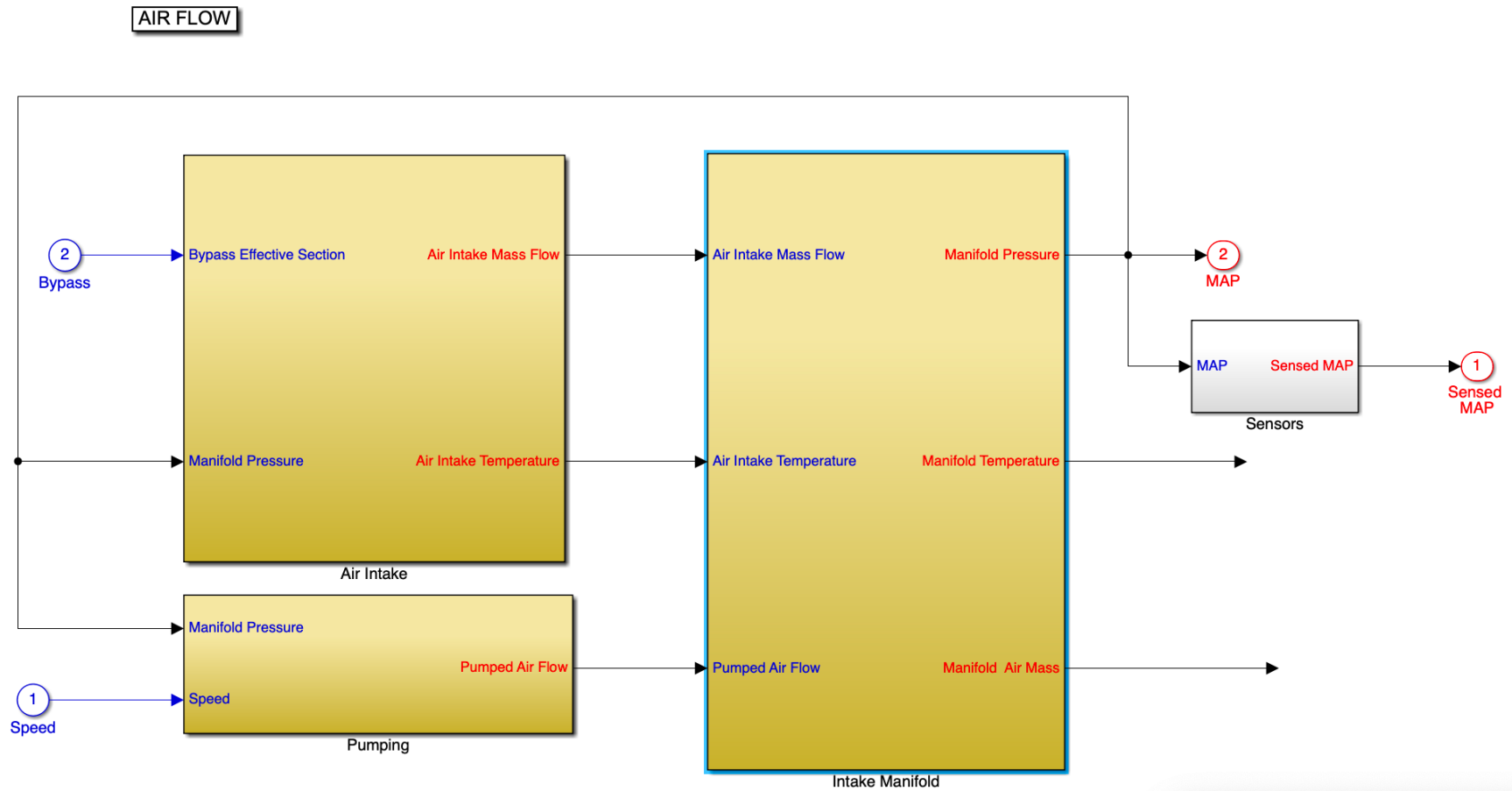
$$f_2(P_{col}) = P_{amb} \sqrt{\frac{2\gamma}{(1-\gamma) \cdot r \cdot T_{amb}} \left[\left(\frac{P_{col}}{P_{amb}} \right)^{\frac{2}{\gamma}} - \left(\frac{P_{col}}{P_{amb}} \right)^{\frac{\gamma+1}{\gamma}} \right]}$$

- ② Sonic mode

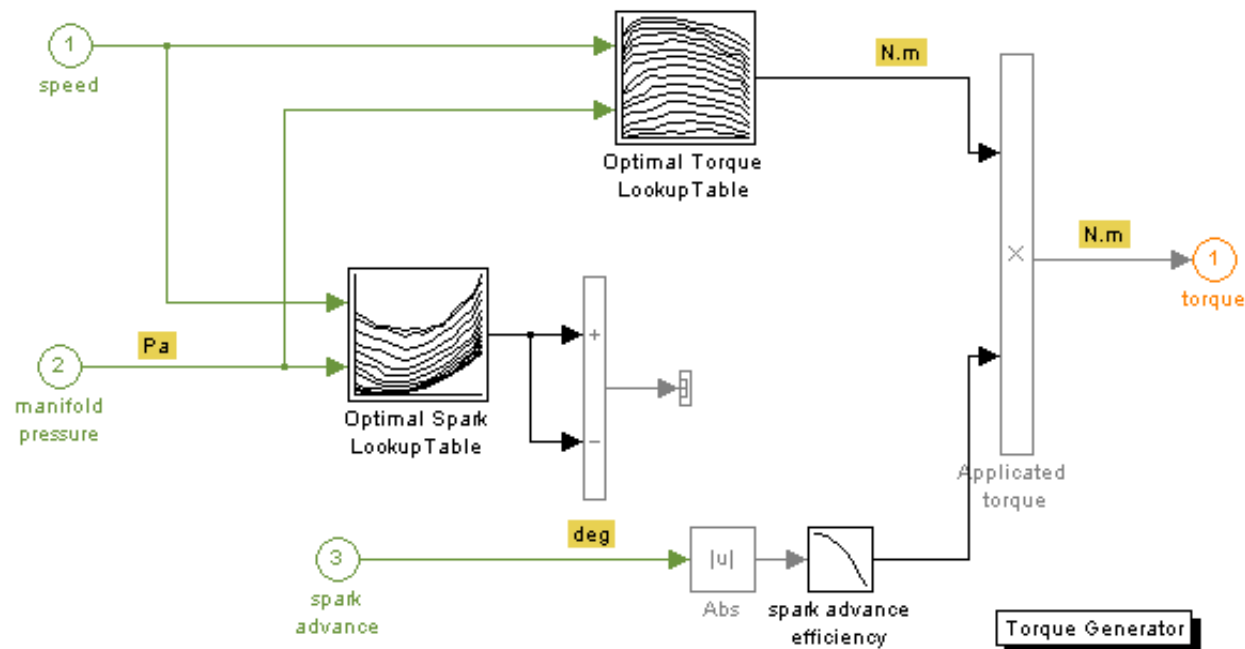
$$\frac{P_{col}}{P_{amb}} < \left(\frac{\gamma + 1}{\gamma} \right)^{\frac{\gamma}{1-\gamma}}$$

$$f_2(P_{col}) = P_{amb} \sqrt{\frac{\gamma}{r \cdot T_{amb}} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{\gamma-1}}}$$

Air Dynamics Model



Dynamics related to the ignition advance

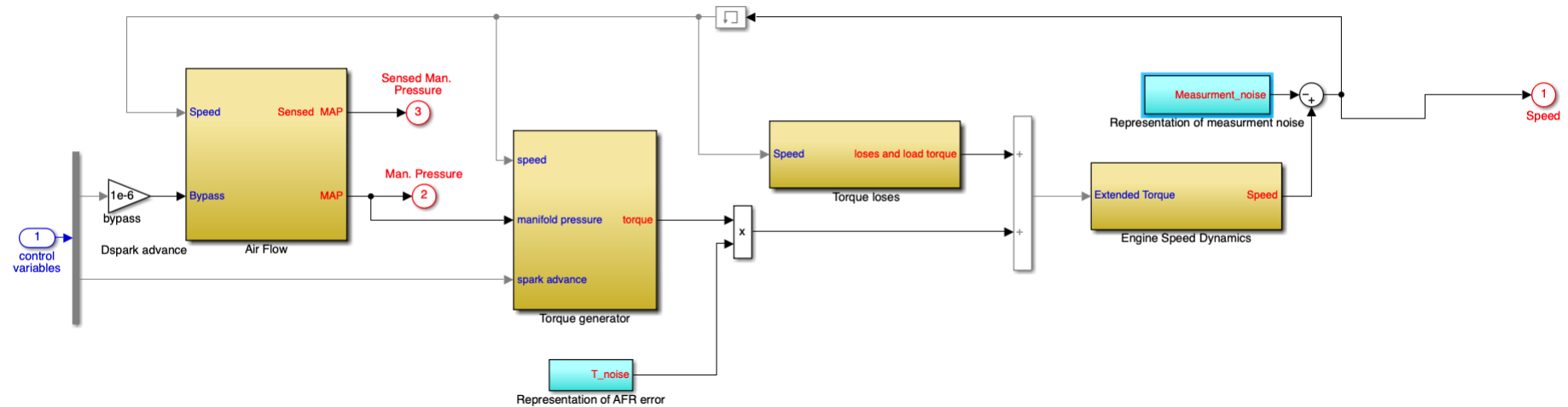


1. No time constant(TDC delay not taken into account)



2. Torque efficiency model

Full model



$$2\pi \cdot J \frac{d(Ne)}{dt} = T_{comb} - T_{load} - T_{frott}$$

Normalisation autour du point de fonctionnement (Ne_0, T_0)

$$2\pi \frac{JNe_0}{T_0} \cdot \frac{d(Ne/Ne_0)}{dt} = \frac{T_{comb}}{T_0} - \frac{T_{load}}{T_0} - K_v \frac{Ne}{Ne_0}$$

$$T_j = 2\pi \frac{JNe_0}{T_0}$$

$$T_{frott} = f(K_v \cdot Ne)$$

$$T_{comb} = \frac{PMI \cdot C_t}{4 \cdot \pi}$$

Models of representation

- Principle
 - Excite the system: What types of signals? Sine? steps? Amplitude?
 - Détermine
 - System Order: 1er ou 2nd order ?
 - Gain,
 - Rising-Time,
 - Response time,
 - Overtaking
- Multiple operating points : 750, 900 et 1000
 - With variation of Spap of +/- 0.5mm² around,
 - With variation of AA +/- 0.5° around 20°.

Closed-loop control (1/4)

- Block diagram of the closed-loop
- Controller Preset from the open-loop: P, PI (PID ?)
 - ! One input at a time,
 - ! Saturation & Integrator.
- Evaluating performance: criteria
- Assessing robustness
 - ! Measurement noises

Closed-loop control (2/4)

- Adjustment PID
 - Pole Placement.
- Evaluating performance: criteria
- Assessing robustness
 - ! Measurement noises
- Compare to previous(s)

Closed-loop control (3/4)

- Frequency decoupling
 - Defining the principle
 - Block diagram
 - What happens to the setting?
- Evaluating performance: criteria
- Assessing robustness
 - ! Measurement noises
- Compare to previous(s)

Closed-loop control (4/4)

- RST or LQ/LQG or Optimization
- Evaluating performance: criteria
- Assessing robustness
 - ! Measurement noises
- Compare to previous(s)